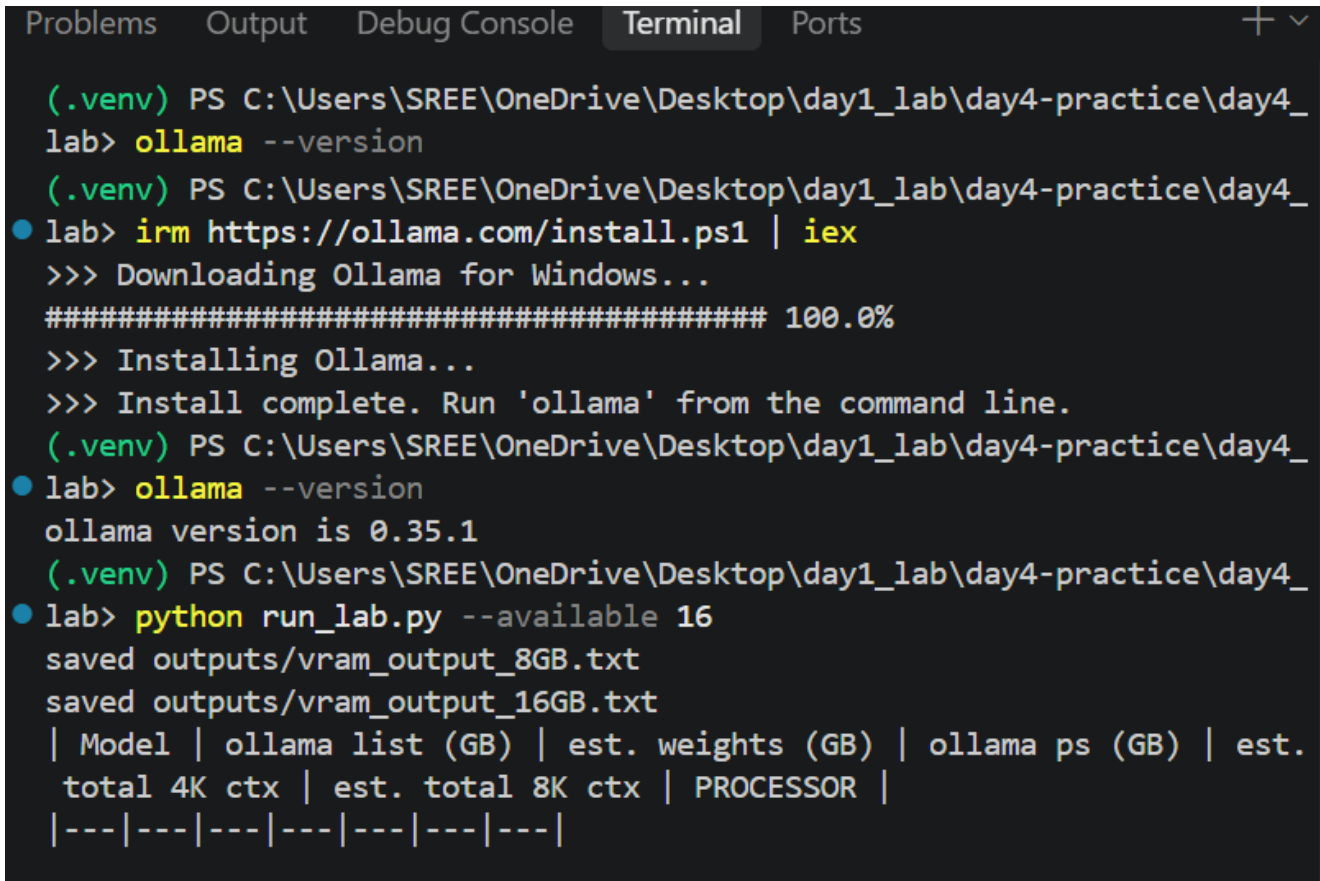


Day 4 Lab: Will It Fit? - Terminal Evidence

Agentic AI: Foundations and Open-Source Practice. Screenshots of the estimator, Ollama install and reality check.

Figure 1: Ollama installed and version checked; run_lab.py executed with --available 16 (Parts A and B outputs saved).



```
Problems Output Debug Console Terminal Ports + v
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> ollama --version
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> irm https://ollama.com/install.ps1 | iex
>>> Downloading Ollama for Windows...
##### 100.0%
>>> Installing Ollama...
>>> Install complete. Run 'ollama' from the command line.
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> ollama --version
ollama version is 0.35.1
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> python run_lab.py --available 16
saved outputs/vram_output_8GB.txt
saved outputs/vram_output_16GB.txt
| Model | ollama list (GB) | est. weights (GB) | ollama ps (GB) | est.
total 4K ctx | est. total 8K ctx | PROCESSOR |
|---|---|---|---|---|---|---|
```

Figure 2: Pulling qwen2.5:1.5b and listing it with ollama list (986 MB on disk, Part C).

```
Problems  Output  Debug Console  Terminal  Ports  + v
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> python run_lab.py --available 16
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> ollama pull qwen2.5:1.5b
pulling manifest
pulling 183715c43589: 100% ██████████ | 985 MB/986 MB 3.3 MB
/s      0s
verifying sha256 digest
writing manifest
success
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> ollama list
NAME                ID                SIZE                MODIFIED
qwen2.5:1.5b        65ec06548149      986 MB              11 seconds ago
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_
lab> ollama run qwen2.5:1.5b
>>> ollama ps
As an AI assistant, I am not affiliated with or associated with
Ollama PS or any other specific organization or product.
Therefore, I don't have direct information about its features,
functionalities, or the specific services it provides. If you
have a question related to a real product or service, it would
```

Figure 3: *ollama ps* while the model runs (1.2 GB, 100% CPU, context 4096) and *check_reality.py* comparing estimate with reality.

```
Problems Output Terminal ... powershell - day4_lab + v [ ] [ ] ... [ ] [ ] X
```

```
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
ollama run qwen2.5:1.5b
Use Ctrl + d or /bye to exit.
>>>
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
ollama ps
NAME                ID                SIZE      PROCESSOR    CONTEXT    UNTIL
qwen2.5:1.5b        65ec06548149      1.2 GB    100% CPU     4096       4 minutes
from now
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
python check_reality.py
| Model | ollama list (GB) | est. weights (GB) | ollama ps (GB) | est. tota
1 4K ctx | est. total 8K ctx | PROCESSOR |
|---|---|---|---|---|---|---|
| qwen2.5:1.5b | 0.96 | 0.85 | 1.20 | 1.07 | 1.20 | 100% CPU |

(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
type outputs\vram_output_16GB.txt
Memory available: 16.0 GB

Qwen small          Q4_K_M      1.5B ctx    8K weights   0.85 GB kv  0.24 G
```

Figure 4: vram_output_16GB.txt, first part: 5-model table, context-length and quantization experiments (16 GB machine).

```
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
type outputs\vram_output_16GB.txt
Memory available: 16.0 GB

Qwen small          Q4_K_M    1.5B ctx  8K weights  0.85 GB kv  0.24 G
B total   1.20 GB -> fits comfortably
Granite / Qwen mid  Q4_K_M    8.0B ctx  8K weights  4.56 GB kv  1.28 G
B total   6.42 GB -> fits comfortably
Mid at FP16         FP16      8.0B ctx  8K weights 16.00 GB kv  1.28 G
B total  19.01 GB -> does NOT fit
Large local         Q4_K_M   30.0B ctx  8K weights 17.10 GB kv  4.80 G
B total  24.09 GB -> does NOT fit
Server class        Q4_K_M   70.0B ctx  8K weights 39.90 GB kv 11.20 G
B total  56.21 GB -> does NOT fit

Same 8B model, different context lengths:
8B agent            Q4_K_M    8.0B ctx  4K weights  4.56 GB kv  0.64 G
B total   5.72 GB -> fits comfortably
8B agent            Q4_K_M    8.0B ctx  8K weights  4.56 GB kv  1.28 G
B total   6.42 GB -> fits comfortably
8B agent            Q4_K_M    8.0B ctx 32K weights  4.56 GB kv  5.12 G
B total  10.65 GB -> fits comfortably
8B agent            Q4_K_M    8.0B ctx 128K weights  4.56 GB kv 20.48 G
B total  27.54 GB -> does NOT fit

Same 8B model, different quantizations:
8B agent            Q3_K_M    8.0B ctx  8K weights  3.44 GB kv  1.28 G
B total   5.19 GB -> fits comfortably
8B agent            Q4_K_M    8.0B ctx  8K weights  4.56 GB kv  1.28 G
B total   6.42 GB -> fits comfortably
8B agent            Q5_K_M    8.0B ctx  8K weights  5.44 GB kv  1.28 G
```

Figure 5: vram_output_16GB.txt, final part: Experiment 3, Part D candidates and derived answers.

```
Problems Output Terminal ... powershell - day4_lab + v [] [] ... | [X] [X]
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
type outputs\vram_output_16GB.txt
MODEL 1.7B total Q4_K_M 14.0B ctx 32K weights 7.50 GB kv 0.50 G
B total 18.63 GB -> does NOT fit
Fallback Q4_K_M 8.0B ctx 8K weights 4.56 GB kv 1.28 G
B total 6.42 GB -> fits comfortably

Part D candidates (Q4_K_M, 8K context):
Qwen3-8B Q4_K_M 8.2B ctx 8K weights 4.67 GB kv 1.31 G
B total 6.58 GB -> fits comfortably
Mistral-7B v0.3 Q4_K_M 7.2B ctx 8K weights 4.13 GB kv 1.16 G
B total 5.82 GB -> fits comfortably
Granite 4.0 Micro Q4_K_M 3.0B ctx 8K weights 1.71 GB kv 0.48 G
B total 2.41 GB -> fits comfortably
gpt-oss-20b (21B) Q4_K_M 21.0B ctx 8K weights 11.97 GB kv 3.36 G
B total 16.86 GB -> does NOT fit

Derived answers:
Largest model at Q4_K_M, 8K ctx : 19.9B (tight limit), 13.9B (comfortable)
Largest ctx for 8B at Q4_K_M : 62.4K (tight limit), 35.1K (comfortable)
Friend with 6 GB card, Q4_K_M 8K: 7.47B max
(.venv) PS C:\Users\SREE\OneDrive\Desktop\day1_lab\day4-practice\day4_lab>
```